

ANSWER KEY — Integration Lesson 40 Test: BC Capstone

For the grader · full working shown · award the point for correct method with a small arithmetic slip · note which tool the student chose

1. Evaluate $\int_1^9 (1/\sqrt{x}) \, dx$. (*all powers*)

Work: rewrite $1/\sqrt{x} = x^{-1/2}$. Antiderivative $= x^{1/2}/(1/2) = 2\sqrt{x}$ (check: $d/dx[2x^{1/2}] = x^{-1/2}$ ✓). Then $[2\sqrt{x}]_1^9 = 2(3) - 2(1) = 6 - 2$. **Answer: 4.** *Classic error: trying the power rule as if the exponent were -1 (that case alone needs $\ln$).*

2. Evaluate $\int_0^2 (3x^2 - 2x) \, dx$. (*Fundamental Theorem*)

Work: $F(x) = x^3 - x^2$ (check: $F' = 3x^2 - 2x$ ✓). $F(2) = 8 - 4 = 4$; $F(0) = 0$. Subtract: $4 - 0$.
Answer: 4.

3. Evaluate $\int_0^1 2x(x^2 + 1)^2 \, dx$. (*u-substitution*)

Work: $u = x^2 + 1$, $du = 2x \, dx$ — the $2x$ is present, so this is a clean substitution. New limits: $x = 0 \rightarrow u = 1$; $x = 1 \rightarrow u = 2$. Integral becomes $\int_1^2 u^2 \, du = [u^3/3]_1^2 = 8/3 - 1/3 = 7/3$. **Answer: 7/3.**
Equally valid without changing limits: the antiderivative is $(x^2+1)^3/3$, and evaluating from $x = 0$ to $x = 1$ gives $8/3 - 1/3 = 7/3$. Classic error: dividing by an extra 2 (writing $(x^2+1)^3/6$, answer $7/6$) — that happens when a student replaces $2x \, dx$ with $du/2$ even though the full $2x$ is already there.

4. Evaluate $\int_0^1 1/(1 + x^2) \, dx$ exactly. (*inverse trig*)

Work: this is the arctan form (no derivative-of-inside present, so u-sub is impossible). $\int 1/(1+x^2) \, dx = \arctan x$. Then $\arctan 1 - \arctan 0 = \pi/4 - 0$. **Answer: $\pi/4$.** *A plain-looking fraction whose area is exactly a quarter of π .*

5. Find $\int x \cos x \, dx$ and check by differentiating. (*integration by parts*)

Work: parts with $u = x$ (simplifies when differentiated), $dv = \cos x \, dx \rightarrow du = dx$, $v = \sin x$. Then $uv - \int v \, du = x \sin x - \int \sin x \, dx = x \sin x + \cos x + C$ (the minus of a minus gives $+$ $\cos x$). **Check:** $d/dx[x \sin x + \cos x] = \sin x + x \cos x - \sin x = x \cos x$ ✓ **Answer: $x \sin x + \cos x + C$.** *Classic error: $x \sin x - \cos x + C$ (sign slip on $\int \sin x = -\cos x$).*

6. Find $\int 1/((x-3)(x+1)) dx$. (*partial fractions*)

Work: write $1/((x-3)(x+1)) = A/(x-3) + B/(x+1) \rightarrow 1 = A(x+1) + B(x-3)$. Plug $x = 3$: $1 = 4A \rightarrow A = 1/4$. Plug $x = -1$: $1 = -4B \rightarrow B = -1/4$. **Verify the split:** $(1/4)[(x+1) - (x-3)]/((x-3)(x+1)) = (1/4)(4)/((x-3)(x+1))$ ✓ Integrate: $(1/4)\ln|x-3| - (1/4)\ln|x+1| + C$. **Answer: $(1/4)\ln|x-3| - (1/4)\ln|x+1| + C$, i.e. $(1/4)\ln|(x-3)/(x+1)| + C$.** Accept either form. Absolute values required.

7. Evaluate $\int_1^\infty 1/x^3 dx$ or show divergence. (*improper integral*)

Work: replace ∞ with b and take a limit. $\int_1^b x^{-3} dx = [-1/(2x^2)]_1^b = -1/(2b^2) + 1/2$. As $b \rightarrow \infty$, $1/(2b^2) \rightarrow 0$, leaving $1/2$. Since $p = 3 > 1$, convergence was expected. **Answer: $1/2$ (convergent).** Full credit requires the limit notation, not plugging ∞ in directly.

8. $g(x) = \int_2^{x^3} \sqrt{1+t} dt$. Find $g'(x)$. (*FTC Part 2*)

Work: FTC Part 2 says $d/dx \int_a^x f(t) dt = f(x)$; when the upper limit is a function, multiply by its derivative (chain rule). Here the upper limit is x^3 , whose derivative is $3x^2$. **Answer: $g'(x) = 3x^2\sqrt{1+x^3}$.** Classic error: answering $\sqrt{1+x^3}$ and forgetting the $\cdot 3x^2$. The lower limit 2 is irrelevant to the derivative.

9. Region under $y = \sqrt{x}$ on $[0, 4]$, revolved about the x -axis. Find the volume. (*volumes — disks*)

Work: disks with radius $\sqrt{x}$, so $V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \int_0^4 x dx = \pi[x^2/2]_0^4 = \pi(8)$. **Answer: 8π .** Squaring the radius first is the key step; a student who integrates $\sqrt{x}$ and then squares gets a wrong answer. Answer must carry π .

10. $v(t) = 2t - 6$ m/s on $[0, 5]$. Find (a) net displacement, (b) total distance. (*net vs total*)

Work: $v = 0$ at $t = 3$; $v < 0$ on $[0, 3]$ (moving backward) and $v > 0$ on $(3, 5]$. Antiderivative $F(t) = t^2 - 6t$; $F(0) = 0$, $F(3) = 9 - 18 = -9$, $F(5) = 25 - 30 = -5$.

(a) Net = $F(5) - F(0) = -5 - 0 = -5$ m (it ends up 5 m *behind* where it started).

(b) Pieces: $[0, 3]$ gives $-9 - 0 = -9$; $[3, 5]$ gives $-5 - (-9) = +4$. Total distance = $|-9| + |4| = 13$ m.

Answer: (a) -5 metres (5 m backward) (b) 13 metres. Classic error: reporting 5 m for both parts, i.e. never splitting at $t = 3$.